

An enriched almost contraction with no enriched-contractive invariant partition

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Abstract

We give a negative answer to the final open problem of arXiv:2103.10289. There is an enriched almost contraction $T : \mathbb{R} \rightarrow \mathbb{R}$ for which no T -invariant partition can have enriched-contractive restrictions; the same example also rules out nonexpansive restrictions. The construction is explicit: start from a two-valued almost contraction S , then undo a Krasnosel'skiĭ average by setting $T = 3S - 2I$. A three-point orbit segment gives an immediate parameter-free contradiction on its forced partition cell.

1 Definitions and interpretation

A self-map A of a normed space is a (δ, L) -almost contraction if

$$\|Ax - Ay\| \leq \delta\|x - y\| + L\|y - Ax\|, \quad 0 < \delta < 1, \quad L \geq 0. \quad (1)$$

A map T is an enriched (b, θ, L) -almost contraction if

$$\|b(x - y) + Tx - Ty\| \leq \theta\|x - y\| + L\|b(x - y) + Tx - y\|, \quad b \geq 0, \quad 0 < \theta < b + 1. \quad (2)$$

It is a (k, η) -enriched contraction if

$$\|k(x - y) + Tx - Ty\| \leq \eta\|x - y\|, \quad k \geq 0, \quad 0 \leq \eta < k + 1. \quad (3)$$

The open problem asks for a partition $X = \bigcup_i X_i$ such that each restriction $T|_{X_i}$ is an enriched contraction (or a nonexpansive map). As in the motivating Picard-operator partitions, “restriction” must mean a self-map: $T(X_i) \subseteq X_i$. Without this invariance, the partition into singletons makes every two-point inequality vacuous, so the question would have a trivial affirmative answer unrelated to its motivation.

2 Construction

Define $S : \mathbb{R} \rightarrow \mathbb{R}$ by

$$Sx = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0, \end{cases} \quad \text{and set} \quad Tx = 3Sx - 2x. \quad (4)$$

Lemma 1. *The step map S is a $(1/2, 2)$ -almost contraction.*

Proof. If x, y lie on the same side of zero, then $Sx = Sy$ and (1) is immediate. If $x < 0 \leq y$, then

$$|Sx - Sy| = 2, \quad \frac{1}{2}|x - y| + 2|y - Sx| = \frac{1}{2}(y - x) + 2(y + 1) \geq 2.$$

If $y < 0 \leq x$, the same calculation gives

$$\frac{1}{2}|x - y| + 2|y - Sx| = \frac{1}{2}(x - y) + 2(1 - y) \geq 2.$$

Thus (1) holds for every ordered pair (x, y) . □

Lemma 2. *The map T in (4) is an enriched $(2, 3/2, 2)$ -almost contraction.*

Proof. The identities

$$2(x - y) + Tx - Ty = 3(Sx - Sy)$$

and

$$2(x - y) + Tx - y = 3(Sx - y)$$

show that (2), with $b = 2$, $\theta = 3/2$, and $L = 2$, is exactly three times the $(1/2, 2)$ -almost-contraction inequality for S . Notice that $3/2 < 3 = b + 1$, as required. Equivalently, the Krasnosel'skiĭ average $(2I + T)/3$ is precisely S . □

3 No admissible invariant partition

Theorem 1. *The enriched almost contraction T from (4) admits neither*

- 1. a T -invariant partition into enriched-contractive restrictions, nor*
- 2. a T -invariant partition into nonexpansive restrictions.*

Consequently, the final open problem of arXiv:2103.10289 has a negative answer.

Proof. The following orbit segment is exact:

$$u := \frac{8}{5} \quad \mapsto \quad v := Tu = -\frac{1}{5} \quad \mapsto \quad Tv = -\frac{13}{5}. \quad (5)$$

Suppose $\{X_i\}$ were a T -invariant partition. The point u belongs to one cell, say X_i , and invariance forces $v = Tu \in X_i$. For this pair,

$$|u - v| = \frac{9}{5}, \quad |Tu - Tv| = \frac{12}{5}. \quad (6)$$

Equation (6) already contradicts nonexpansiveness on X_i .

Now suppose that $T|_{X_i}$ were a (k, η) -enriched contraction, where $k \geq 0$ and $\eta < k + 1$. Both differences in (6) have the same sign before absolute values, so applying (3) to u, v gives

$$\frac{9k + 12}{5} = |k(u - v) + Tu - Tv| \leq \eta|u - v| = \frac{9\eta}{5}.$$

Hence

$$\eta \geq k + \frac{4}{3} > k + 1,$$

contrary to the defining condition $\eta < k + 1$. Thus the forced cell is neither enriched contractive nor nonexpansive, for every possible choice of its parameters. □

4 Why the construction is robust

The mechanism is not tied to the displayed numbers. If S is any almost contraction and $b \geq 0$, then

$$T = (b + 1)S - bI$$

is an enriched almost contraction, because $(bI + T)/(b + 1) = S$. The target question concerns invariant pieces for T , not for its averaged map S . De-averaging can create a positive expanding secant along a T -orbit even though S has the standard almost-contractive partition properties. Such a secant is incompatible with (3), since in one dimension every positive secant slope of an enriched contraction must be strictly less than one. Formula (5) realizes this obstruction with slope $4/3$.

References

- [1] V. Berinde and M. Păcurar, *Krasnoselskij-type algorithms for variational inequality problems and fixed point problems in Banach spaces*, arXiv:2103.10289.